Conference on Empirical Methods in Natural Language Processing, pp. 2086–2099, 2024.

Hendrycks, D., Burns, C., Kadavath, S., Arora, A., Basart, S., Tang, E., Song, D., and Steinhardt, J. Measuring mathematical problem solving with the math dataset. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2)*, 2021.

Huang, K., Guo, J., Li, Z., Ji, X., Ge, J., Li, W., Guo, Y., Cai, T., Yuan, H., Wang, R., et al. Math-perturb: Benchmarking llms' math reasoning abilities against hard perturbations. In *Forty-second International Conference on Machine Learning*, 2025.

Jiang, A. Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D. S., de Las Casas, D., Bressand, F., Lengyel, G., Lample, G., Saulnier, L., Lavaud, L. R., Lachaux, M.-A., Stock, P., Scao, T. L., Lavril, T., Wang, T., and Sayed, W. E. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.

Kojima, T., Gu, S. S., Reid, M., Matsuo, Y., and Iwasawa, Y. Large language models are zero-shot reasoners. *Advances in neural information processing systems*, 35: 22199–22213, 2022.

Li, Y. Policy guided tree search for enhanced llm reasoning. In *Forty-second International Conference on Machine Learning*, 2025.

Lightman, H., Kosaraju, V., Burda, Y., Edwards, H., Baker, B., Lee, T., Leike, J., Schulman, J., Sutskever, I., and Cobbe, K. Let's verify step by step. In *The Twelfth International Conference on Learning Representations*, 2023.

Lin, Z., Liang, T., Xu, J., Liu, Q., Wang, X., Luo, R., Shi, C., Li, S., Yang, Y., and Tu, Z. Critical tokens matter: Token-level contrastive estimation enhances llm's reasoning capability. In *Forty-second International Conference on Machine Learning*, 2025.

Ling, Z., Fang, Y., Li, X., Huang, Z., Lee, M., Memisevic, R., and Su, H. Deductive verification of chain-of-thought reasoning. *Advances in Neural Information Processing Systems*, 36:36407–36433, 2023.

Ma, C., Zhao, H., Zhang, J., He, J., and Kong, L. Non-myopic generation of language models for reasoning and planning. In *The Thirteenth International Conference on Learning Representations*, 2025.

Madaan, A., Tandon, N., Gupta, P., Hallinan, S., Gao, L., Wiegreffe, S., Alon, U., Dziri, N., Prabhumoye, S., Yang, Y., et al. Self-refine: Iterative refinement with self-feedback. *Advances in Neural Information Processing Systems*, 36:46534–46594, 2023.

Miao, N., Teh, Y. W., and Rainforth, T. Selfcheck: Using llms to zero-shot check their own step-by-step reasoning. In *The Twelfth International Conference on Learning Representations*, 2024.

Patel, A., Bhattamishra, S., and Goyal, N. Are nlp models really able to solve simple math word problems? In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 2080–2094, 2021.

Qi, Z., Mingyuan, M., Xu, J., Zhang, L. L., Yang, F., and Yang, M. Mutual reasoning makes smaller llms stronger problem-solver. In *The Thirteenth International Conference on Learning Representations*, 2025.

Snell, C., Lee, J., Xu, K., and Kumar, A. Scaling llm test-time compute optimally can be more effective than scaling model parameters. *arXiv preprint*, 2024.

Sprague, Z. R., Yin, F., Rodriguez, J. D., Jiang, D., Wadhwa, M., Singhal, P., Zhao, X., Ye, X., Mahowald, K., and Durrett, G. To cot or not to cot? chain-of-thought helps mainly on math and symbolic reasoning. In *The Thirteenth International Conference on Learning Representations*, 2025.

Stechly, K., Valmeekam, K., and Kambhampati, S. On the self-verification limitations of large language models on reasoning and planning tasks. In *The Thirteenth International Conference on Learning Representations*, 2025.

Tanneru, S. H., Agarwal, C., and Lakkaraju, H. Quantifying uncertainty in natural language explanations of large language models. In *International Conference on Artificial Intelligence and Statistics*, pp. 1072–1080. PMLR, 2024.

Tyen, G., Mansoor, H., Cărbune, V., Chen, Y. P., and Mak, T. Llms cannot find reasoning errors, but can correct them given the error location. In *Findings of the Association for Computational Linguistics ACL 2024*, pp. 13894–13908, 2024.

Uesato, J., Kushman, N., Kumar, R., Song, F., Siegel, N., Wang, L., Creswell, A., Irving, G., and Higgins, I. Solving math word problems with process-and outcome-based feedback. *arXiv preprint*, 2022.

Wan, G., Wu, Y., Chen, J., and Li, S. Reasoning aware self-consistency: Leveraging reasoning paths for efficient llm sampling. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 3613–3635, 2025.

Wang, P., Li, L., Shao, Z., Xu, R., Dai, D., Li, Y., Chen, D., Wu, Y., and Sui, Z. Math-shepherd: Verify and reinforce llms step-by-step without human annotations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 9426–9439, 2024.

Wang, X., Wei, J., Schuurmans, D., Le, Q. V., Chi, E. H., Narang, S., Chowdhery, A., and Zhou, D. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations*, 2023.

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q. V., Zhou, D., et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.

Wu, Y., Sun, Z., Li, S., Welleck, S., and Yang, Y. Inference scaling laws: An empirical analysis of compute-optimal inference for llm problem-solving. In *The Thirteenth International Conference on Learning Representations*, 2025.

Wu, Z., Zeng, Q., Zhang, Z., Tan, Z., Shen, C., and Jiang, M. Large language models can self-correct with key condition verification. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 12846–12867, 2024.

Xie, Y., Kawaguchi, K., Zhao, Y., Zhao, J. X., Kan, M.-Y., He, J., and Xie, M. Self-evaluation guided beam search for reasoning. *Advances in Neural Information Processing Systems*, 36:41618–41650, 2023.

Xu, F., Yan, H., Ma, C., Zhao, H., Liu, J., Lin, Q., and Wu, Z. Phi-decoding: Adaptive foresight sampling for balanced inference-time exploration and exploitation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 13214–13227, 2025.

Yadav, V., Tang, Z., and Srinivasan, V. Pag-llm: Paraphrase and aggregate with large language models for minimizing intent classification errors. In *Proceedings of the 47th international ACM SIGIR conference on research and development in information retrieval*, pp. 2569–2573, 2024.

Yang, A., Li, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Gao, C., Huang, C., Lv, C., et al. Qwen3 technical report. *arXiv preprint*, 2025a.

Yang, L., Yu, Z., Zhang, T., Xu, M., Gonzalez, J. E., CUI, B., and YAN, S. Supercorrect: Advancing small llm reasoning with thought template distillation and self-correction. In *The Thirteenth International Conference on Learning Representations*, 2025b.

Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T., Cao, Y., and Narasimhan, K. Tree of thoughts: Deliberate problem solving with large language models. *Advances in neural information processing systems*, 36:11809–11822, 2023.

Yin, Z., Sun, Q., Guo, Q., Zeng, Z., Li, X., Dai, J., Cheng, Q., Huang, X.-J., and Qiu, X. Reasoning in flux: Enhancing large language models reasoning through uncertainty-aware adaptive guidance. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 2401–2416, 2024.

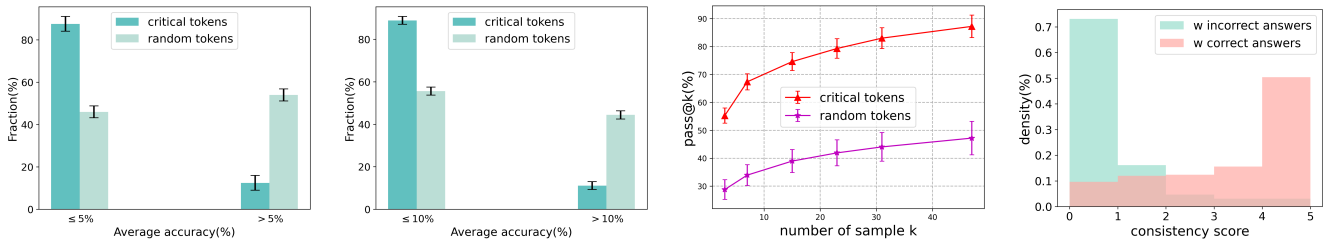
Yuan, L., Li, W., Chen, H., Cui, G., Ding, N., Zhang, K., Zhou, B., Liu, Z., and Peng, H. Free process rewards without process labels. In *Forty-second International Conference on Machine Learning*, 2025.

Zhang, D., Zhoubian, S., Hu, Z., Yue, Y., Dong, Y., and Tang, J. Rest-mcts*: Llm self-training via process reward guided tree search. *Advances in Neural Information Processing Systems*, 37:64735–64772, 2024.

Zhang, Z., Zhang, A., Li, M., and Smola, A. Automatic chain of thought prompting in large language models. In *The Eleventh International Conference on Learning Representations*, 2022.

Zhou, Y., Muresanu, A. I., Han, Z., Paster, K., Pitis, S., Chan, H., and Ba, J. Large language models are human-level prompt engineers. In *The eleventh international conference on learning representations*, 2022.

Zhou, Y., Zhu, Y., Antognini, D., Kim, Y., and Zhang, Y. Paraphrase and solve: Exploring and exploiting the impact of surface form on mathematical reasoning in large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 2793–2804, 2024.



(a) *(b)* *(c)* *(d)*

Figure 8. (a) Comparison of the fraction of tokens whose associated rollout correctness falls below or above 5%, for tokens extracted by our method versus randomly selected tokens. (b) Comparison of the fraction of the subsequent tokens whose associated rollout correctness falls below or above 10%, for tokens extracted by our method versus randomly selected tokens. (c) Comparison of the impact of extracted tokens by our method versus random tokens on LLM reasoning performance. (d) Comparison of the density distributions of consistency scores for rollouts with correct and incorrect answers. All the evaluations are conducted on Math500 dataset.

A. Empirical Study

we follow the previous pipeline to conduct a quantitative analysis to examine the authenticity and impact of tokens extracted via paraphrastic probing, comparing them against randomly selected tokens. Specifically, we sample 100 instances with incorrect reasoning trajectories from the Math500 (Hendrycks et al., 2021) training sets. Following the paraphrastic probing pipeline, we identify candidate critical tokens in the early reasoning steps. In each run, we randomly sample 40% of the candidate critical tokens for evaluation and repeat the experiment independently 10 times. For comparison, we apply the same procedure to randomly selected tokens. All evaluations are conducted using Llama-3.1-8B-Instruct (Grattafiori et al., 2024). We first compute the average accuracy of rollouts generated from partial trajectories truncated at the position of the extracted tokens, and compare these results with those obtained from random-token truncation. As shown in Figure 8a, we also obverse a large proportion of the extracted tokens exhibit near-zero correctness. We further evaluate the average accuracy of rollouts initiated from subsequent tokens (5–10 tokens). The results in Figure 8b also indicate that errors persist beyond the identified positions. Finally, we substitute the extracted critical tokens with alternative tokens and roll out multiple new reasoning trajectories. As shown in Figure 8c, replacing critical tokens leads to a significantly larger improvement in reasoning accuracy compared to substituting random tokens.

To justify our paraphrase consistency mechanism on Math500 dataset (Hendrycks et al., 2021), we follow the same procedure. The evaluation is conducted on Llama-3.1-8B-Instruct (Grattafiori et al., 2024). We then analyze the distribution of consistency scores for rollouts that yield correct versus incorrect answers. As shown in Figure 8d, majority of rollouts with correct answers achieve a consistency score of at least 1, whereas this proportion drops sharply for rollouts with incorrect answers.

B. Automatic Prompt Engineering for Paraphrasing

The quality of paraphrased questions is crucial for our framework, as it directly affects both the identification of critical tokens and the stability of paraphrase-based consistency verification. To ensure high-quality paraphrasing, we adopt Automatic Prompt Engineering (APE) (Zhou et al., 2022) as an evaluation to optimize the paraphrasing instruction. This allows us to systematically control the quality of paraphrases rather than relying solely on manually written prompts. The APE procedure we employ consists of four steps:

- We create a small set of original questions paired with valid paraphrased variants. Using this exemplar set, the model generates multiple candidate instructions that could produce the paraphrased outputs from the original questions.
- Prompt each candidate instruction to the language model to generate paraphrases for the original problems and compare the mean solve rate change before and after paraphrasing.
- Choose the instruction that maximizes the mean solve rate change.
- Repeat the previous steps multiple times.

In our experiments, we initialize the paraphrasing instruction and iteratively refine it using APE with samples drawn from the GSM8K and Math500 training sets. With this approach, the refined paraphrasing prompt helps us reliably produce